THE EULER CHARACTERISTIC IS UNCHANGED BY EXTENSION OF THE GROUND FIELD

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Flat base change for proper schemes over field extensions. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology. The identification of cohomology groups.

Source reference. Original reference 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k'/k be an extension of fields. Let X be a proper scheme over k. Let $\mathcal{F}$ be a coherent sheaf on X. Let $\mathcal{F}'$ be the pullback of $\mathcal{F}$ to $X_{k'}$. Then $\chi(X, \mathcal{F}) = \chi(X', \mathcal{F}')$.*

Proof. This is true because

$$H^i(X_{k'}, \mathcal{F}') = H^i(X, \mathcal{F}) \otimes_k k'$$

by flat base change, see Cohomology of Schemes, Lemma 02KH. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.